Trader Behaviour & Market Sentiment Analysis

Insights from Trade-Level PnL, Leverage, Win Rate, and Sentiment Data

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1. Introduction

This report provides an in depth analysis of trader behavior and market sentiment using trade level data. By examining Closed PnL, leverage, trade count, win rate, and sentiment value, the goal is to understand how emotional market conditions influence trading performance, risk taking, and decision making.

Objective: The analysis aims to connect trader behavior, market sentiment, and profitability, providing insights into risk management, opportunity exploitation, and emotional trading patterns.

Scope of Analysis:

- Identify behavioral patterns and potential contrarian strategies.
- Assess trader participation across different sentiment phases.

Dataset Description:

- `Closed PnL` **– Realized profit or loss per trade**
- `Leverage` **– Risk exposure of each trade**
- `Side` **– BUY or SELL trade**
- `Classification` **– Market sentiment label (Extreme Fear, Fear, Greed, Extreme Greed, Neutral)**
- `Timestamp` **– Time of trade execution**

- `value` – **Numeric representation of sentiment**

2. Trading Activity by Sentiment

This section analyzes trade counts and participation based on market sentiment. We will interpret behavioral trends observed from the data.

Finding 1: Peak Trading During Moderate Fear

- The Fear phase records the highest trade count (24,749 trades).
- Traders actively exploit market uncertainty and short term volatility.

Finding 2: Extreme Fear suppresses participation

- Extreme Fear shows the lowest participation (7,715 trades).
- Risk aversion dominates.

3. Profitability (Closed PnL) Analysis

This section examines the distribution of profits and losses across different market sentiment phases, using mean, median, standard deviation, skewness, and kurtosis.

	classification	Closed PnL		Leverage
		mean	sum	mean
0	Extreme Fear	11.922975	91985.749998	8.538533
1	Extreme Greed	13.146128	221919.794657	8.198061
2	Fear	13.054170	323077.651059	8.431268
3	Greed	10.056115	206743.676064	8.018690
4	Neutral	10.398961	159187.287763	8.533274

Descriptive Statistics:

	Closed PnL
count	85212.000000
mean	11.769635
std	22.671218
min	-56.206644
25%	0.312800
50%	3.838475
75%	17.542936
max	94.849796

dtype: float64
Skewness: 1.2826393897797876
Kurtosis: 2.434751785159885

Finding 1: Highest Mean PnL During Extreme Greed and Fear

- Mean Closed PnL is highest in Extreme Greed (13.15) and Fear (13.05) phases
- Traders capitalize on volatility in Fear, while overconfident or trend-following strategies dominate in Extreme Greed.
- Standard deviation is higher for Extreme Fear (23.65) and Extreme Greed (21.43), reflecting wide variation in trade outcomes

Finding 2: Narrower PnL Spread in Neutral and Greed Phases

- Lower spread and standard deviation in Neutral (21.68) and Greed (23.60) phases.
- Market stability reduces extreme outcomes, leading to more predictable but lower
- magnitude profits.

4. Win Rate Analysis

This section evaluates the success rate of trades (Win Rate) across different market sentiment phases and examines the relationship between sentiment and trader performance

	classification	Win_Rate
0	Extreme Fear	0.800778
1	Extreme Greed	0.886381
2	Fear	0.880561
3	Greed	0.762245
4	Neutral	0.829174

Finding 1: Highest Win Rates During Extreme Greed and Fear

- Extreme Greed: 88.6% win rate
- High volatility and strong market trends provide opportunities for profitable trades, particularly for experienced or risk-tolerant traders.

Finding 2: Acting opposite to the prevailing sentiment improves success

- SELL trades often outperform BUY trades during fearful phases, aligning with contrarian trading strategies.

Finding 3: Moderate Win Rates During Neutral Sentiment

- Neutral phase has a win rate of 82.9%, lower than Extreme Greed or Fear.
- Reflects stable market conditions with fewer large movements.

5. Leverage Analysis

Leverage reflects the risk exposure taken by traders on each trade. This section examines how leverage varies across market sentiment phases and its impact on profitability and risk.

```
classification
Extreme Fear     8.538533
Extreme Greed    8.198061
Fear             8.431268
Greed            8.018690
Neutral          8.533274
```

Finding 1: Extreme Fear encourages Aggressive leverage

- Average leverage peaks at 8.54 during Extreme Fear.
- Suggests that traders who remain active in panic phases take larger positions, potentially to capitalize on market dislocations.

Finding 2: Extreme Fear suppresses participation

- Extreme Fear shows the lowest participation (7,715 trades).
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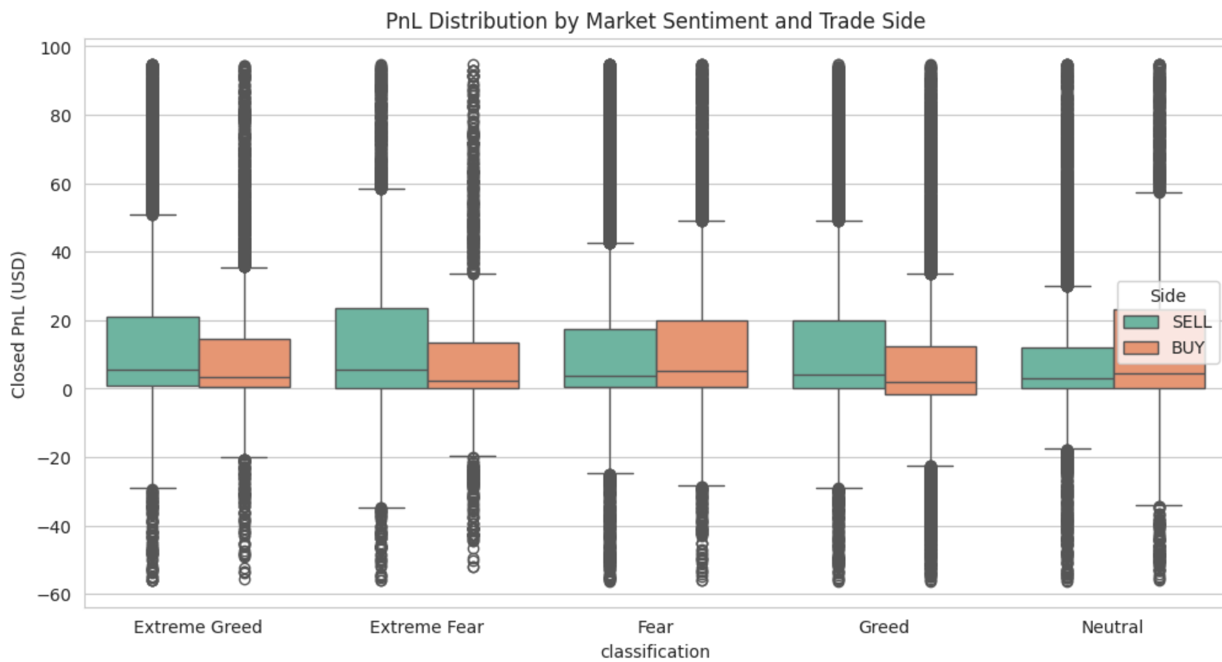
6. BUY vs SELL Performance

This section compares the performance of BUY and SELL trades, analyzing differences in profitability and highlighting potential contrarian trading opportunities.



Finding 1: Extreme Fear Favors SELL Trades

- Fearful markets create opportunities for short-selling or downside positions.
- Suggests traders can benefit from contrarian strategies in panic phases.

Finding 2: BUY trades may underperform during Greed.

- Suggests that acting opposite to prevailing market sentiment (contrarian strategy) can enhance returns.

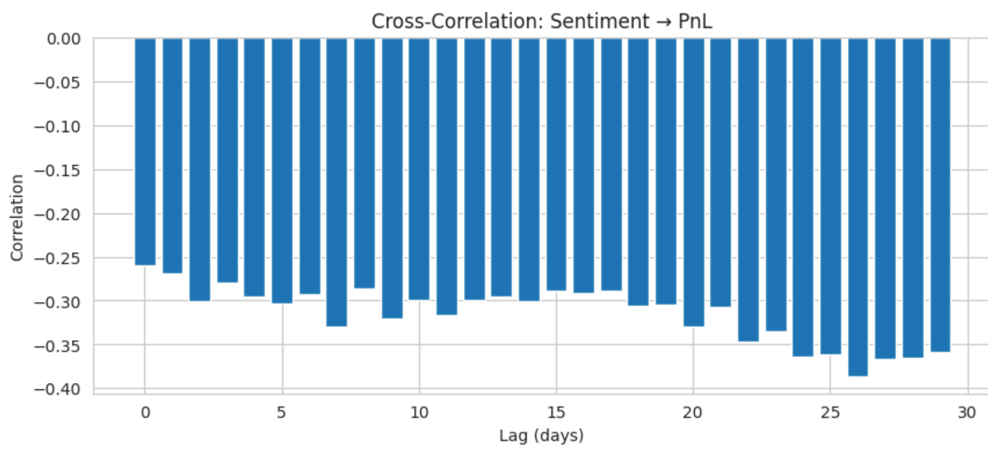
Finding 3: SELL Trades Outperform BUY Trades Overall

- A t-test comparing BUY vs SELL Closed PnL shows a significant difference ($t = -11.05$, $p \approx 2.37 \times 10^{-28}$).
- SELL trades have higher median PnL than BUY trades across most sentiment phases.

7. Correlation Analysis

Correlation analysis examines the relationships between Leverage, Closed PnL, and Sentiment Value, helping understand how risk exposure and emotional sentiment influence trade outcomes.

	Leverage	Closed PnL	value
Leverage	1.000000	0.101374	-0.057707
Closed PnL	0.101374	1.000000	-0.004250
value	-0.057707	-0.004250	1.000000



Finding 1: Leverage Slightly Negatively Correlated with Sentiment Value

- Correlation coefficient ≈ −0.06.
- Traders tend to increase leverage in fearful markets and reduce it in overly optimistic markets.

Finding 2: Minimal Direct Correlation Between Sentiment Value and PnL.

- Correlation coefficient ≈ −0.004.
- Numeric sentiment alone is not a strong predictor of profitability

8. Regression Analysis (Predicting PnL)

A linear regression model was used to examine how Leverage and numeric Sentiment Value predict Closed PnL. This helps determine the strength of these factors in explaining trade profitability.

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Regression Coefficients: [0.88749275 0.00123749]
Intercept: 4.34760130193673
R² Score: 0.010, MSE: 513.70
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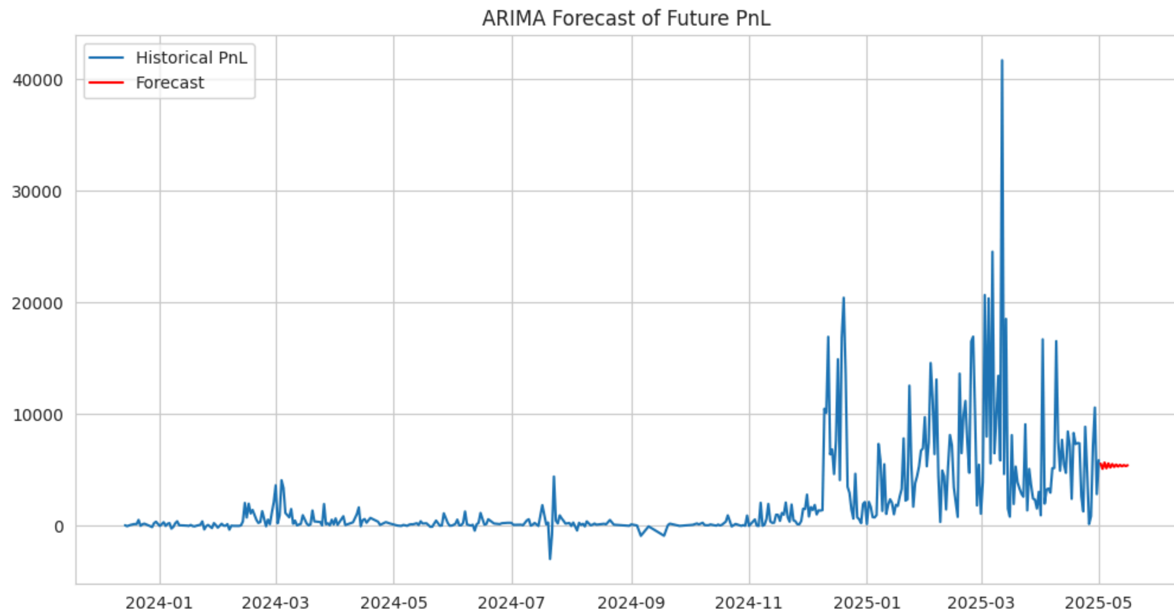


Finding 1: Low Predictive Power of Leverage and Sentiment

- The model yielded an $R^2 \approx 0.013$, meaning only ~1% of PnL variance is explained by leverage and sentiment.
- Indicates that other factors (trade timing, strategy, market volatility) play a much larger role in determining profitability.

9. Behavioral & Market Insights

This section synthesizes behavioral patterns observed in trading data, connecting market sentiment, trader behavior, and profitability to principles of behavioral finance.

Finding 1: High Volatility During Extreme Greed and Extreme Fear

- Traders take larger positions and use higher leverage during extreme sentiment phases.
- Indicates that emotional extremes drive aggressive trading, leading to wide PnL variability.

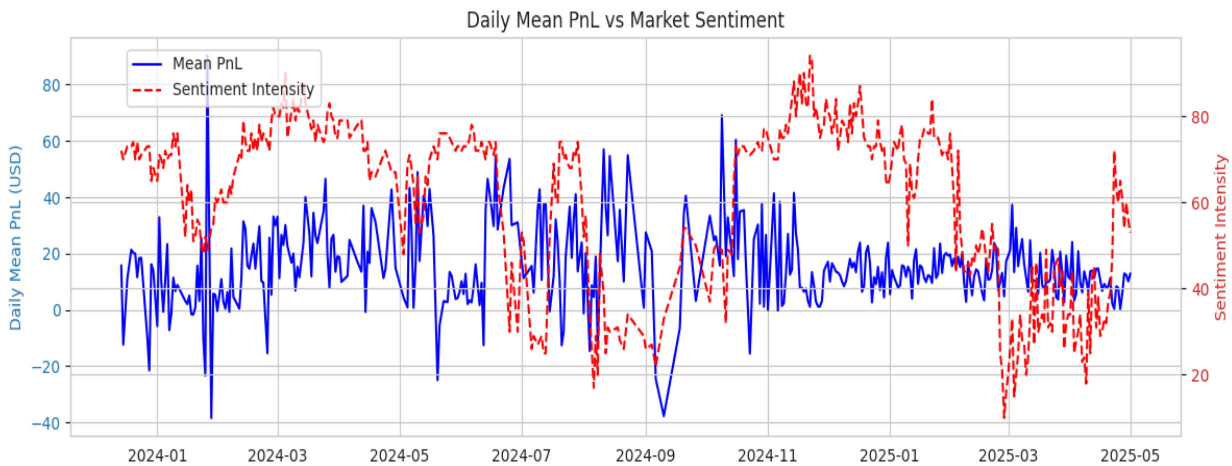
Finding 2: Contrarian Trading Strategies Yield Better Returns

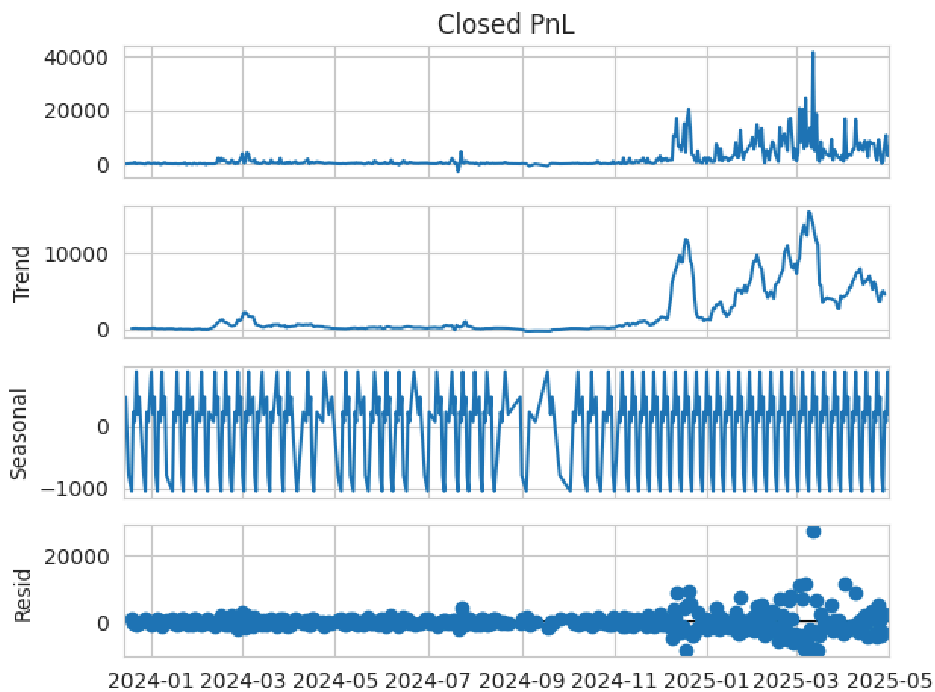
- Traders acting opposite to prevailing sentiment achieve higher profitability.

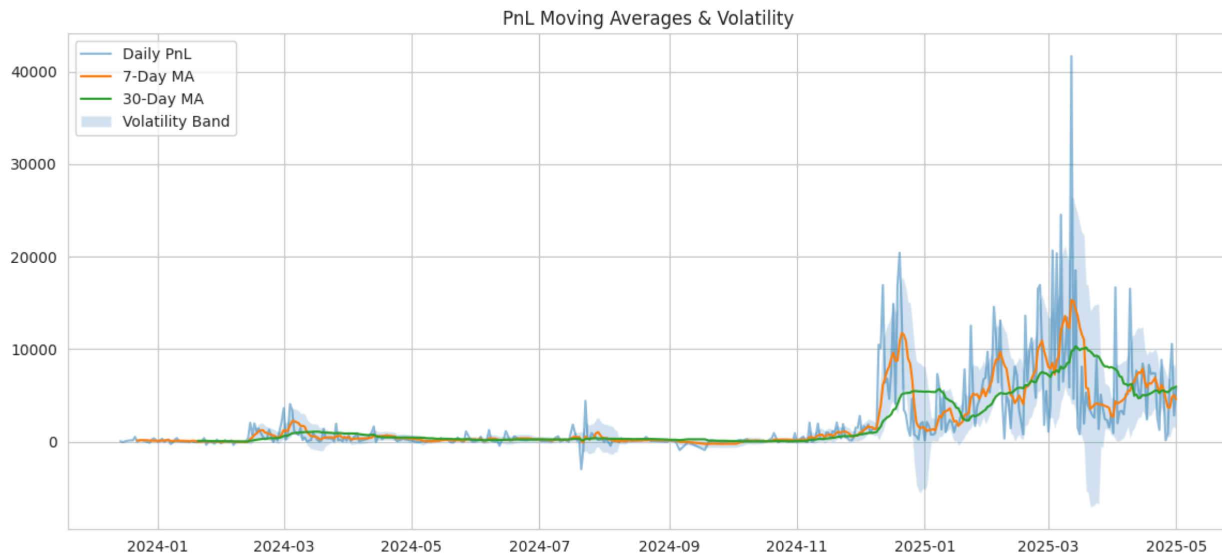
Finding 3: Median PnL Remains Positive Across Sentiments

- Most trades yield moderate profits, even in volatile conditions.
- While outliers exist, consistent risk management ensures overall profitability.

10. Time Series Analysis







Finding 1: High-frequency seasonal pattern is present in the PnL data

- This pattern likely corresponds to a weekly trading cycle, where certain days of the week systematically exhibit different PnL characteristics
- Amplitude of this seasonal effect also appears to increase in late 2024 and early 2025

Finding 2: Regime shift occurred in trading activity in late 2024.

- This indicates fundamental change in trading strategy, scale of operations, or market conditions.

11 . Summary

1 . **Emotional Extremes Influence Risk and Profitability**
2 . **Contrarian Strategies are Rewarding**
3 . **Moderate Sentiment Phases Provide Stable Returns**
4 . **Importance of Risk Management**

12 . Recommendations for Traders.

1 . **Exploit Fear-Driven Opportunities:** Focus on contrarian trades during fearful market phases.

2 . **Manage Leverage Wisely:** Use higher leverage selectively in volatile markets and reduce it in calm, optimistic markets.

3 . Prioritize Risk Management: Protect against extreme losses by diversifying positions and setting stop-losses.

4 . Avoid Overconfidence in Greedy Markets: Optimistic phases can create complacency; exercise caution.

5. Focus on Timing and Strategy: Sentiment alone does not guarantee profits; success depends on market timing and disciplined execution.

12. Conclusion

Trader behavior is strongly influenced by market sentiment, particularly emotional extremes. Profitable trading is supported by contrarian strategies, careful leverage management, and risk aware execution. Markets with moderate sentiment provide stable opportunities, while extreme sentiment phases offer high risk, high-reward potential for disciplined traders.

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